[myKomuniti: Public Hazard Reporting & Prevention System]

PROJECT PROPOSAL & PROJECT CHARTER

1.0 Project Overview

1.1 Project title and operational domain

Project Title:
myKomuniti: Public Hazard Reporting & Prevention System.

SDG:
SDG 11 – Sustainable Cities and Communities.
Target 11.5 – Reduce the adverse effects of natural disasters.

Operational Domain:
myKomuniti is a web-based public hazard reporting and monitoring system designed for the residents of Taman Bertam Jaya, Melaka.

The system focuses on enabling residents to report potential public hazards in their neighbourhood, including:

- Flooding and flood-prone areas
- Blocked drains
- Fallen trees or dangerous branches
- Damaged roads
- Unstable slopes
- Other potential public hazards

Residents can submit hazard reports containing the hazard category, description, location, severity level and photo evidence. Community administrators can review and manage submitted reports, update report statuses, record follow-up actions and monitor hazard trends through statistics and reports. When a hazard requires official action, the community administrators can refer it to the relevant local authority.

The system aims to improve community awareness, organise hazard information and support communication and follow-up between residents and the relevant parties to help create a safer neighbourhood.

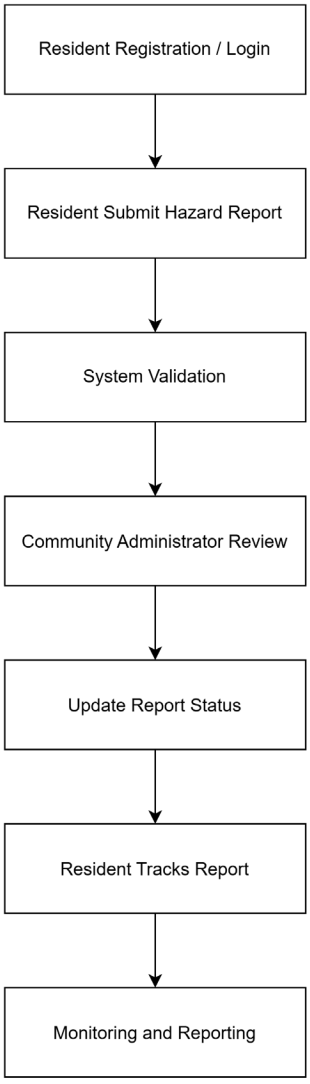
1.2 Background and existing workflow

Residents may encounter public hazards in their neighbourhood, particularly during heavy rain, storms and other situations that may increase the risk of hazards.

In the proposed case study at Taman Bertam Jaya, Melaka, residents may report public hazards through different communication channels. However, hazard information may not always include complete details, such as the exact location, severity and photo, making it difficult to organise and monitor reports effectively.

The proposed myKomuniti system aims to provide a centralised platform for residents to submit structured hazard reports. Community administrators can review reports, update their statuses and record follow-up actions. The system aims to improve the organisation, monitoring and follow-up of public hazard reports, helping support a safer neighbourhood.

Proposed Workflow:



1.3 Case study feasibility

Criterion	Evidence / justification
Authentic stakeholder domain	The proposed system focuses on public hazard reporting and monitoring within the residents' community of Taman Bertam Jaya, Melaka. The group will gather information through stakeholder interviews and observations to understand residents' needs and current reporting practices.
Automated data lifecycle	The system manages hazard reports from submission to closure. Each report contains details such as hazard category, location, severity, description, submission date, status, follow-up actions and resolution date.
Feasible within 14 weeks	The system will be developed using PHP, MySQL, HTML, CSS and JavaScript. The scope is limited to hazard reporting, report management, monitoring and PDF reporting.
Avoids anti-patterns	The project is a transactional web-based system that includes user roles, CRUD operations, relational database tables, report status management, search and filtering, and reporting features. Its scope is designed to be manageable within the 14-week development period.

2.0 Problem Statements and Objectives

ID	Problem statement (current reality, pain point, impact)	Corresponding objective (what the system will achieve)
PS1 / O1	Residents may lack a centralised platform to submit complete public hazard information, including category, location, severity and photo evidence. This may make hazard reports difficult to organise and monitor.	To develop a web-based platform that allows residents to submit structured public hazard reports.
PS2 / O2	Residents may have limited visibility of the progress of their submitted reports after reporting a hazard.	To allow residents to track the status of their own hazard reports.
PS3 / O3	Multiple reports may be submitted for the same hazard or location, making it difficult to identify repeated reports.	To implement a mechanism for identifying and managing repeated hazard reports based on location and hazard category.
PS4 / O4 (optional)	Hazard information such as location, category, severity and status may be difficult to summarize for monitoring purposes.	To provide an administrative dashboard containing hazard statistics and operational summaries.

2.1 Project goal and intended outcomes

Project Goal

To develop a web-based public hazard reporting and monitoring system that enables residents of Taman Bertam Jaya, Melaka, to report public hazards and allows community administrators to review, manage, track and monitor reported hazards.

Intended Outcomes

By Week 14, the system is expected to:

1. Allow residents and community administrators to log in securely according to their assigned roles.
2. Allow residents to submit hazard reports containing the category, description, location, severity and photo evidence.
3. Restrict residents to viewing and managing only their own reports, according to the system's permissions.
4. Allow community administrators to view and manage submitted hazard reports.
5. Provide a clear report status workflow from submission to closure.
6. Help community administrators identify potential duplicate reports based on hazard category and location.
7. Calculate report processing and resolution times using report timestamps.
8. Display hazard statistics by location, category, severity and status.
9. Generate PDF reports for administrative monitoring and record-keeping.
10. Store system data in a relational MySQL database containing at least five related tables.
11. Protect user passwords using secure password hashing.

3.0 Scope Baseline and System Boundaries

Domain	In scope – committed this semester	Out of scope – deferred or excluded
User roles and access	Resident and Community Administrators with different permissions	Additional external roles
Authentication	Registration, login, logout, session management and password hashing	Google/Facebook login, biometric authentication and SSO
Hazard reporting	Create, view, update and controlled cancellation of resident's own reports	Direct emergency dispatch
Hazard management	Community Administrators review reports, update statuses and record follow-up actions.	Automatic physical repair/maintenance
Duplicate reports	Identify similar active reports using location and hazard category	AI-based duplicate detection
Location	Submit the hazard location byentering street name / house address within Taman Bertam Jaya.	Live GPS tracking
Photo upload	JPG/JPEG/PNG files, maximum 5 MB per image and maximum 3 images per report	Video uploads
Data and reporting	Relational MySQL database, filters, statistics and operational summaries	Advanced AI analytics
Resolution monitoring	Track submitted, reviewed, verified, action and resolution dates	Real-time government workflow integration
PDF reporting	Generate PDF summaries of hazard reports	Automated email/SMS distribution
Security	Password hashing and role-based access	Advanced penetration testing
Notifications	Display report status and updates within the system.	Paid SMS/WhatsApp APIs

3.1 Proposed core modules

Module	Main functions	Primary role(s)
M01 User Management	Registration, login, logout, profile management, password hashing and role management	Resident and Community Administrator
M02 Hazard Reporting	Submit hazard report, upload photos, select category/location/severity and view own reports	Resident
M03 Hazard Management & Tracking	Review reports, verify reports, update status, record action taken and manage repeated reports	Community Administrator
M04 Monitoring & Analytics	Search/filter reports, view statistics, calculate resolution time and generate PDF reports	Community Administrator

3.2 Assumptions, constraints and dependencies

Category	Detail / mitigation
Assumptions	Residents have access to a web browser and an internet connection. The residents' community can provide relevant information about local hazards, reporting needs and suitable hazard categories. These assumptions will be verified through stakeholder interviews where possible.
Constraints	Project must be completed within 14 weeks. Development is limited by team skills, time and available laboratory resources.
Dependencies / risks	Access to stakeholder information, database hosting, PHP/MySQL environment and sample hazard data. **Risk:** 1. Stakeholder unavailable - Use approved observation/public information and clearly document assumptions if direct interview is unavailable. 2. File upload problems - Apply file type and size validation before storing images. 3. Security issues - Use password hashing, prepared SQL statements, sessions and role-based access control. 4. Scope becoming too large - Keep live GPS, SMS, AI and external APIs outside the current scope.

4.0 Project Charter and Team Organisation

4.1 Team structure

Role	Assigned member	Main accountability
Project Manager (PM)	Deanna Anessa Binti Muhamad Zaharin (D032510191)	Schedule, scope, communication, team coordination and final submission
Lead Systems Analyst & UI/UX Designer	Muhammad Ihtisyam Fadlullah Bin Faisal (D032510552)	Stakeholder interviews, requirements, use cases and interface design
Lead Full-Stack / Database Engineer	Amir Hamzah Bin Mohd Hanif (D032510148)	Architecture, relational schema, backend/frontend integration, CRUD
Lead Full-Stack / Database Engineer	Ahmad Alhaziq Deen Bin Ahmad Ronnie Eddly (D032510362)	Architecture, relational schema, backend/frontend integration, CRUD
QA & DevOps Engineer	Aliah Suffiah Binti Azahari (D032510035)	Test planning, evidence, repository quality and documentation

4.2 RACI responsibility matrix

R = Responsible (does the work); A = Accountable (final owner/sign-off); C = Consulted; I = Informed. R/A means the same role does and owns the task.

Deliverable / activity	PM	Analyst	Developer	QA
Stakeholder interview protocol	A	R	C	I
Project Proposal & Charter	R/A	C	C	C
SRS and CRUD mapping	I	R/A	C	C
Database and codebase	I	C	R/A	C
Test cases and QA log	I	C	C	R/A
Final integration / demonstration	A	R	R	R

4.3 Working agreement

Agreement area	Group commitment
Meeting frequency and channel	Group discussion at least once per week through physical/online meetings and WhatsApp communication.
Version control and documentation	Use GitHub for source code and maintain clear commit messages. Major changes should be reviewed by another team member.
Task assignment and progress	Tasks will be assigned through a WhatsApp group and shared document and reviewed during weekly meetings.
Conflict and absence management	Members must inform the group when unavailable. Unfinished tasks will be reassigned when necessary.
Quality and sign-off	Completed features must be tested before being considered complete.

5.0 Milestone Schedule and Verification

Gate	Target	Verifiable deliverable	Owner
M1	Week 2	Project proposal, stakeholder/domain confirmation, scope and requirements	PM
M2	Week 5	Stakeholder interview findings, Software Requirements Specification (SRS), functional and non-functional requirements, use case diagram	Analyst
M3	Week 7	Figma wireframes and navigation flow, system architecture, database ERD and database schema	UI/UX & DB lead
M4	Week 11	Functional website with registration/login, hazard reporting, report management and tracking, MySQL database integration	Developer
M5	Week 13	Test plan, test cases, test results, bug-fix record and validated system features	QA
M6	Week 14	Completed website, final documentation, PDF report functionality, presentation slides and demonstration	Entire team

5.1 Week 1–2 immediate actions

Immediate actions:

1. Confirm the selected stakeholder: Residents' Community (Komuniti Penduduk).
2. Confirm the implementation area: Taman Bertam Jaya, Melaka.
3. Conduct a stakeholder interview or observation where possible.
4. Identify the current public hazard reporting workflow.
5. Confirm system users and permissions: Residents and Community Administrators.
6. Finalise the problem statements based on stakeholder findings.
7. Finalise the project objectives.
8. Define the report status workflow, from submission to closure.
9. Define the method for identifying and managing potential duplicate reports.
10. Confirm photo upload restrictions, including file formats, file size and maximum number of images.
11. Review the revised project proposal with the supervisor or lecturer.

6.0 Charter Approval / Sign-off

Person	Name / signature	Date
Project Manager	[Deanna Anessa / signature]	[22/10/2026]
Group member representatives	[Muhammad Ihtisyam / signatures]	[22/10/2026]
Group member representatives	[Amir Hamzah / signatures]	[22/10/2026]
Group member representatives	[Ahmad Alhaziq / signatures]	[22/10/2026]
Group member representatives	[Aliah Suffiah / signatures]	[22/10/2026]
Supervisor / lecturer	[Approval / comments]	[22/10/2026]

Appendix A. Evidence and references

Stakeholder Interview

